

The zeroLET Task Model and its Application to Offset Design Space Exploration

Abstract—Chains of asynchronous tasks are nowadays widespread in a broad range of applications, most notably in automotive. In asynchronous communication, tasks do not block on the availability of data. Rather they are released by periodic timers and then read/write data upon the start/completion of jobs. LET (Logical Execution Time) tasks were conceived to decouple the communication instants from the scheduling logic. In LET tasks, read and write of data always occur at predetermined instants. The analysis of chains of LET tasks does depend on the offset of tasks, i.e., the read/write instants of the first job of any task. This opens to the problem of a proper offset assignment in task chains.

In this paper, we propose a few contributions to address this problem. First, we propose the zeroLET task model, an equivalent simpler model that decouples the per-task and synchronization contributions to end-to-end latencies. We exploit such a novel model to precisely determine the equivalence classes of offsets with the same reaction time. We use this result to find the maximal/minimal reaction time among all possible offset, and we link it to existing results. Finally, we experimentally evaluate the run-time of offset design space exploration.

Index Terms—Task chains, LET tasks, task offsets, real-time systems, end-to-end latency, reaction time, offset equivalence, offset design.

I. INTRODUCTION

Autonomous driving systems [1], whether built on ROS [2] or not, execute ordered modular tasks mapped to ECUs [3], forming *task chains*. Communication within a chain is either:

- *synchronous*, when tasks are triggered by the availability of new data, or
- *asynchronous*, when tasks read/write data with non-blocking primitives, and are then triggered by periodic timers.

In these chains, it is then relevant to study end-to-end latency metrics such as *reaction time* and *data age*, widely studied by the community and adopted in AUTOSAR standard [4].

Logical Execution Time (LET) emerged as useful abstraction to decouple the communication instants from the task schedule. The LET communication paradigm specifies the instants when data is read and written. The abstraction of LET communication guarantees determinism and provides robustness for functional expansion and changes [5]. Conversely, LET also increases the end-to-end latency in the task chain [4]. Most existing analysis methods for LET depend on explicit knowledge of task offsets, which are the specific times when the first instance (job) of any task performs read and write operations. In industrial deployments, these offsets are frequently opaque to the end-users (due to the lack of accessible APIs). The precise offset values are also unavailable or ill-defined in practical systems (due to clock drift). Especially, in architectures with complex task chain interactions,

the lack of this information complicates the analysis process, potentially leading to pessimistic or erroneous latency bounds. Some researchers have focused on this issue. Martinez et al. [6] studied the selection of task phases and obtained a set of offsets that might reduce the end-to-end latency, but still require explicit offset information. Günzel and Becker [5] studied the task phase selection in semi-harmonic systems to minimize the end-to-end latency of the task chain. However, such approach relies on a specific periodic structure.

To address this problem, we propose the novel *zeroLET task model*, an equivalent LET task model, which allows a simpler analysis by separating the end-to-end latency in its intra-task and inter-task components. Then, we exploit the zeroLET model to determine equivalent offset assignments that yield the same reaction time. Such observation, allows a drastic reduction of the offset design space.

Contributions of this paper: After introducing the standard task and chain model, Section III defines the *maximum/minimum offset-free MRT*, which are the longest/shortest maximum reaction time (MRT) under all possible offset combinations. Section IV introduces the zeroLET task model, which allows decoupling the intra-task latency and the inter-task latency. A more efficient exploration of the offset design space is enabled by the zeroLET model, as demonstrated in Section V. Finally in Section VI, we exploit the zeroLET model to evaluate all possible offsets for some selected choice of periods and we address the scalability of the offset space exploration. Our evaluation confirms the validity of the demonstrated bound (Theorem 2) as well as the offset design rules for max-harmonic tasks [5].

II. RELATED WORK

The analysis of task chains has received considerable attention in real-time systems, particularly in automotive applications. Davare et al. [7] proposed a period optimization method to minimize the maximum reaction time while preserving the single processor schedulability. Durr et al. [8] proposed the concept of forward/backward job chains. Based on this concept, they proposed a tighter bound by exploiting task scheduling information. Günzel et al. [9] determined an upper bound to the reaction time by dividing task chains. Then in [10], Günzel et al. provide a formal analysis framework for end-to-end cause-effect chains with different types of tasks and communication mechanisms. Similarly, Becker et al. [11] presented methods for calculating end-to-end latency based on different levels of system information.

In combination with real-time scheduling strategies, Kloda et al. [12] analyzed periodic task chains under fixed-priority preemptive scheduling on a single processor. Ren et al. [13]

analyzed periodic task chains under partitioned fixed-priority preemptive scheduling on multiprocessor systems. Additionally, job-level dependencies have also been considered in task-chain analysis [14]. In contrast, event-triggered task chains have also been studied, in which each task release is triggered by an output signal from its predecessor. For such synchronous communication task chains, Tang et al. [15] have focused on the impact of buffer size on synchronization mechanisms and end-to-end latency.

Such studies often rely on task scheduling behavior, LET specifies communication instants independently of the execution schedule. Kordon et al. [16] used the LET model for maximum data age analysis of periodic task chains. Hamann et al. [4] analyzed reaction times for sporadic task chains. In [17], Bini et al. proposed eliminating jitter in LET periodic chains by copying tasks. Wang et al. [18] proposed a new event model that explicitly captures uncertainty in read/write operations. Paladino et al. [19] proposed to optimize priorities to reduce the reaction time. Pazzaglia et al. focused on LET performance optimization [20], [21]. Becker et al. [11] studied the case in which timing information is increasingly precise with the deployment process.

Several extensions to the standard LET model have also been investigated. The flexible LET (fLET) model optimizes read and write operations to improve end-to-end timing performance by introducing virtual offsets and deadlines [22]. Gemlau et al. [23] proposed a system-level LET (SLLET) and comprehensively evaluated its applicability to complex embedded hardware/software systems.

A notable family of related works addressed the development of synchronous languages for embedded systems. Esterel [24] is probably credited as the first language addressing real-time synchronization. Lustre [25] introduced a declarative synchronous data-flow language for real-time reactive systems. Signal allowed multiple clocks [26]. The above languages evolved from research prototypes into industrial technologies [27]. In a very related work [28], it was proposed an offset assignment, which fulfills a given communication pattern avoiding synchronization primitives.

Existing research on end-to-end analysis of task chains, especially LET models, commonly assume that task configuration parameters such as offsets are known. Martinez et al. [6] examined the impact of offsets on latency and investigated how task phases can be selected to reduce end-to-end latency. Schlatow [29] proposed a mixed-integer linear programming optimization method to select multiple configurations, including offsets, to minimize data age. Pellizzoni and Lipari [30] proved that allocating offsets based on EDF scheduling can bring performance advantages. Schliecker and Ernst [31] proposed a recursive method to compute tight end-to-end latencies in heterogeneous multiprocessor systems by accounting for pipelining and transient effects, whereas this work focuses on latency bounds for specific job sequences to derive job-level dependencies. Luo et al. [32] modeled external events with finite validity intervals and analyzed how the minimum validity interval of external events guarantees response time. Günzel and Becker [5] focused on determining task offset for semi-harmonic systems and optimizing the

minimization of end-to-end latency of the task chain.

The above studies focus on the selection of task stages, and some can obtain a set of offsets that may reduce end-to-end latency. However, most researchers require predefined initial offset information rather than providing a universal upper bound applicable to arbitrary offsets. In this paper, we propose the zeroLET model, which simplifies the analysis while preserving equivalence in evaluating end-to-end latency, and enables the determination of latency variation under different offsets without explicitly requiring offset information.

III. TASKS AND CHAINS: MODEL

We analyze a single *chain* of tasks, modeled by a sequence $\mathcal{T} = [\tau_1, \dots, \tau_n]$ of n tasks. Each task, denoted by τ_i , releases a job every *period* $T_i \in \mathbb{N}$. We assume that the indices of jobs belong to $\mathbb{Z}$, the set of all integers, including the negative numbers. In fact, as observed by Günzel et al. [33], the maximum reaction time (a metric of great interest in task chains [34]) should not be evaluated over the jobs belonging to the *warm-up period*. Hence, considering the job indices in $\mathbb{Z}$ eliminates any transient effect and the necessity to drop jobs over the warm-up period. Furthermore, we emphasize that the reaction time during the system start-up time is normally of less interest than the reaction time during the full operational stage.

Tasks communicate by means of shared memory. Each job $j \in \mathbb{Z}$ of τ_i performs the following operations, in this order:

- 1) it reads the input data from shared memory at the *read instant* $\text{rd}_{\tau_i}(j)$;
- 2) it performs its computations; and
- 3) it writes the output data to shared memory at the *write instant* $\text{wr}_{\tau_i}(j)$, making it available to other tasks.

In the chain $\mathcal{T}$, the communication follows the order of tasks: for any $i = 2, \dots, n$, the task τ_{i-1} writes some data that is read by τ_i . The first task τ_1 reads data from the external environment, and the last task τ_n writes its output to the external environment.

Following previous works [35], [36], [20], we use a generalization of the Logical Execution Time (LET) paradigm to implement inter-task communication. In this generalization, data reads and data writes are performed at fixed instants, but not necessarily at the start and end of the task periods. We allow any protocol for managing shared memory, as long as (i) read/write operations are non-blocking, (ii) the data is available to the consumer(s) only when the producer task has completed the write operation, and (iii) each write operation by the same task overwrites the previously written data, thus read operations will only access the last written data. Note that these requirements hold in AUTOSAR implementations of the LET paradigm [37]. The fixed instants specified by the read and write offsets can be enforced by using GMF/LET communication tasks and RTE-generated code.

Under such a generalized LET model, the read/write instants of the jobs of τ_i (enumerated with index j) are written as

$$\text{rd}_{\tau_i}(j) = j T_i + \phi_{\tau_i}^r, \quad \text{wr}_{\tau_i}(j) = j T_i + \phi_{\tau_i}^w, \quad (1)$$

with:

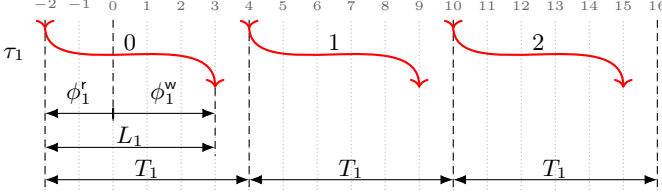


Fig. 1. Representing the communication instants of the LET task $\tau_1 = \langle 6, -2, 3 \rangle$. Each job is represented by an arrow starting from the read instant and ending at the write instant. Job indices are next to the arrows (jobs 0, 1, and 2 represented in the figure). According to Eq. (1): $\text{rd}_1(0) = -2$, $\text{rd}_1(1) = 4, \dots$ and $\text{wr}_1(0) = 3$, $\text{wr}_1(1) = 9$, and so on.

- $\phi_{\tau_i}^r$ representing the read offset of τ_i ,
- $\phi_{\tau_i}^w$ representing the write offset of τ_i .

The job j of task τ_i can then start executing at any time not earlier than $\text{rd}_{\tau_i}(j)$ and complete at any time not later than $\text{wr}_{\tau_i}(j)$. Both in this paper and in the related literature, it is convenient to define the task latency as

$$L_i = \phi_{\tau_i}^w - \phi_{\tau_i}^r, \quad (2)$$

which represents the time elapsing from the read to the write instants of task τ_i . In our LET model, such a latency is constant among the jobs, whereas in different communication schemes such as implicit communication it is not the case [7].

The offsets $\phi_{\tau_i}^r$ and $\phi_{\tau_i}^w$, also called *task offsets* and *task deadlines* in other works, may take any value in $\mathbb{Z}$, provided that causality holds, (i.e., $\phi_{\tau_i}^w \geq \phi_{\tau_i}^r$ or $L_i \geq 0$). Our definition allows negative values for the offsets. We remark that the “time 0” has no special interpretation in this paper (such as the typical “critical instant” of the system start-up time), i.e., having the read instant at negative time does not invalidate our contribution. Since we are only interested in how task τ_i communicates with the others, we represent it by the tuple $\tau_i = \langle T_i, \phi_{\tau_i}^r, \phi_{\tau_i}^w \rangle$. Under the original definition of LET [38], [39], all tasks are $\tau_i = \langle T_i, 0, T_i \rangle$, with reads and writes occurring at the beginning and end of the tasks’ periods, respectively. This work hence considers a more general interpretation of LET where the read and write phases are arbitrary, but anyway constant. We underline that the generality offered by arbitrary read/write offset enables a broader exploration of possible task schedules. Figure 1 shows an example of the communication instants of a LET task.

Like tasks, the chain $\mathcal{T}$ has jobs too. A job chain $(j_1, \dots, j_n) \in \mathbb{Z}^n$ is a tuple of jobs of tasks $\tau_i \in \mathcal{T}$ such that for any $i = 1, \dots, n-1$, the job j_{i+1} is the earliest job of τ_{i+1} that reads the data written by the job j_i of task τ_i . Formally, the above requirement for job j_{i+1} is translated by

$$\begin{aligned} j_{i+1} &= \min\{j \in \mathbb{Z} : \text{wr}_{\tau_i}(j_i) \leq \text{rd}_{\tau_{i+1}}(j)\} \\ j_{i+1} &= \min\{j \in \mathbb{Z} : j_i T_i + \phi_{\tau_i}^w \leq j T_{i+1} + \phi_{\tau_{i+1}}^r\} \\ j_{i+1} &= \left\lceil \frac{j_i T_i + \phi_{\tau_i}^w - \phi_{\tau_{i+1}}^r}{T_{i+1}} \right\rceil. \end{aligned}$$

Some authors use the term *immediate forward job chain* [8], [9], [33] for the job chain $(j_1, \dots, j_n)$, as for each job j_i starting from job j_1 of τ_1 , the job chain contains the job j_{i+1} reading immediately after j_i .

TABLE I
SUMMARY OF NOTATIONS, WITH SECTION OF FIRST APPEARANCE OF THE SYMBOL.

Symbol	Interpretation	Sec.
τ_i	Task	III
$\mathcal{T}$	Chain of tasks	III
$n = \mathcal{T} $	Number of tasks	III
T_i	Period of task τ_i	III
H_i	$\text{lcm}(T_1, \dots, T_i)$, if $i = n$ hyperperiod	III, V
G_i	$\text{gcd}(H_{i-1}, T_i)$, $i \geq 2$	V
$\text{rd}_{\tau_i}(j)$	Read instant of job j of τ_i	III
$\text{wr}_{\tau_i}(j)$	Write instant of job j of τ_i	III
$\phi_{\tau_i}^r$	Read offset of τ_i	III
$\phi_{\tau_i}^w$	Write offset of τ_i	III
L_i	Latency of τ_i	III
$(j_1, \dots, j_n)$	Job chain	III
$L_{\mathcal{T}}(t_0, \phi_{\tau_i}^r, \phi_{\tau_i}^w)$	Reaction time for $\mathcal{T}$ from t_0	III
$L_{\mathcal{T}}(\phi_{\tau_i}^r, \phi_{\tau_i}^w)$	Maximum reaction time (MRT) for $\mathcal{T}$	III
$L_{\mathcal{T}}^+$	Maximum offset-free MRT	III
$L_{\mathcal{T}}^-$	Minimum offset-free MRT	III
ζ_i	zeroLET task	IV
$\mathcal{Z}$	Chain of zeroLET tasks	IV
ϕ_{ζ_i}	Offset of ζ_i (both read and write)	IV

A typical metric of interest in chains of tasks is the reaction time. Informally speaking, the reaction time is the time elapsed from an external event occurring at some instant t_0 sensed by the first task of the chain τ_1 until its effect is propagates to the last task τ_n .

Definition 1 (Reaction time). *Given an external event occurring at time $t_0 \in \mathbb{Z}$ and a chain $\mathcal{T}$, we define the reaction time of chain $\mathcal{T}$ to the event occurring at t_0 by*

$$L_{\mathcal{T}}(t_0, \phi_{\tau_i}^r, \phi_{\tau_i}^w) = \text{wr}_{\tau_n}(j_n) - t_0, \quad (3)$$

with j_n defined by

$$j_1 = \left\lceil \frac{t_0 - \phi_{\tau_1}^r}{T_1} \right\rceil, \quad j_{i+1} = \left\lceil \frac{j_i T_i + \phi_{\tau_i}^w - \phi_{\tau_{i+1}}^r}{T_{i+1}} \right\rceil. \quad (4)$$

We remark that the borrowed original definition [9] was considering the time t_0 of the external event belonging to the real numbers $\mathbb{R}$. For homogeneity with all parameters, we prefer to keep t_0 integer as this difference does not affect the contributions of this paper. When commenting our results, it will be our care to remark how the same result would extend to the case of real-valued parameters.

The notation for the reaction time $L_{\mathcal{T}}(t_0, \phi_{\tau_i}^r, \phi_{\tau_i}^w)$ above explicitly indicates its dependence on the read and write offsets of all tasks, $(\phi_{\tau_1}^r, \phi_{\tau_1}^w, \dots, \phi_{\tau_n}^r, \phi_{\tau_n}^w)$, and this complete offset assignment is compactly denoted by $(\phi_{\tau_i}^r, \phi_{\tau_i}^w)$, where $i = 1, \dots, n$.

By using the above definition, it is possible to define the maximum reaction time.

Definition 2 (Maximum reaction time (MRT)). *Given a task chain $\mathcal{T}$, we define maximum reaction time (MRT) of the chain*

$$L_{\mathcal{T}}(\phi_{\tau_i}^r, \phi_{\tau_i}^w) = \sup_{t_0 \in \mathbb{Z}} L_{\mathcal{T}}(t_0, \phi_{\tau_i}^r, \phi_{\tau_i}^w). \quad (5)$$

In words, the MRT $L_{\mathcal{T}}$ measures the longest waiting time from some event occurring at t_0 and its reaction at $\text{wr}_{\tau_n}(j_n)$,

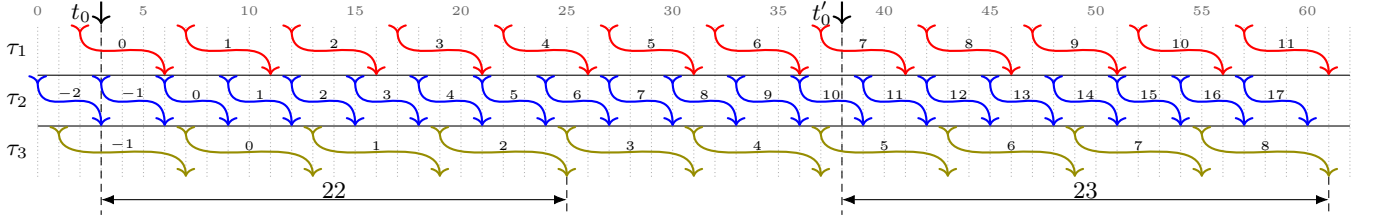


Fig. 2. Showing an example of task chain $\mathcal{T} = [\tau_1, \tau_2, \tau_3]$ with $\tau_1 = \langle 5, 2, 6 \rangle$, $\tau_2 = \langle 3, 6, 9 \rangle$, $\tau_3 = \langle 6, 7, 13 \rangle$. As stated in Eq. (4), the job chain following from an external event occurring at $t_0 = 3$ (a small black arrow pointing down), is $(1, 2, 2)$ with reaction time $L_{\mathcal{T}}(3, \phi_{\tau_1}^r, \phi_{\tau_2}^w) = \text{wr}_{\tau_3}(2) - 3 = 25 - 3 = 22$. If instead the external event is at $t'_0 = 38$ then the job chain from it is $(8, 14, 8)$ with reaction time $L_{\mathcal{T}}(38, \phi_{\tau_1}^r, \phi_{\tau_2}^w) = 61 - 38 = 23$. Such a value is also the MRT $L_{\mathcal{T}}(\phi_{\tau_1}^r, \phi_{\tau_2}^w)$ for these offsets, as defined in Eq. (5). However, for these task periods of $(5, 3, 6)$, the maximum offset-free MRT is $L_{\mathcal{T}}^+ = 24$, and the read offsets $(\phi_{\tau_1}^r, \phi_{\tau_2}^r, \phi_{\tau_3}^r) = (0, 0, 2)$ is one possible assignment that can achieve this value.

with j_n being the job of τ_n which is first affected by the event, as computed from Eq. (4).

A standard observation for the choice of t_0 , that maximizes $L_{\mathcal{T}}(t_0, \phi_{\tau_i}^r, \phi_{\tau_i}^w)$ in Eq. (5), is that t_0 should be one time unit after some read instant of τ_1 . This is apparent from Eq. (3): reducing t_0 without changing any of the j_i of (4) certainly increases $L_{\mathcal{T}}(t_0, \phi_{\tau_i}^r, \phi_{\tau_i}^w)$. Formally, we can restrict the supremum of (5) to $t_0 = (j_1 - 1)T_1 + \phi_{\tau_1}^r + 1 = \text{rd}_{\tau_1}(j_1) - T_1 + 1$ for some j_1 , meaning that we can compute $L_{\mathcal{T}}(\phi_{\tau_i}^r, \phi_{\tau_i}^w)$ by

$$L_{\mathcal{T}}(\phi_{\tau_i}^r, \phi_{\tau_i}^w) = \sup_{j_1 \in \mathbb{Z}} \{ \text{wr}_{\tau_n}(j_n) - \text{rd}_{\tau_1}(j_1) \} + T_1 - 1. \quad (6)$$

In Eq. (6) above, and throughout the whole paper, we write j_n to mean the calculation of j_n from $n - 1$ applications of the ceiling function of Eq. (4) to the job index j_1 of task τ_1 . Hence, j_n and all job indices in the job chain $(j_1, \dots, j_n)$ do depend on j_1 despite the notation not representing such a dependency explicitly. We remark that such a notation is widely adopted in the related literature.

The pattern of read and write instants repeats every $H_n = \text{lcm}(T_1, \dots, T_n)$, called *hyperperiod* of the tasks in the chain. This implies that the supremum of (6) with $j_1 \in \mathbb{Z}$, may be restricted to $j_1 \in [0, \frac{H_n}{T_1} - 1]$ only. In this case, the supremum becomes a maximum as it is made over a finite set. This means that $L_{\mathcal{T}}(\phi_{\tau_i}^r, \phi_{\tau_i}^w)$ can be more practically computed by

$$L_{\mathcal{T}}(\phi_{\tau_i}^r, \phi_{\tau_i}^w) = \max_{j_1 \in [0, \frac{H_n}{T_1} - 1]} \{ \text{wr}_{\tau_n}(j_n) - \text{rd}_{\tau_1}(j_1) \} + T_1 - 1. \quad (7)$$

Finally, as stated in the motivation for this work, the task offsets may be unknown or difficult to set precisely. In this case, however, we assume that the task latencies L_i may not vary. In fact, it is reasonable to assume that the start times of tasks are not synchronized. However, write and read instants of the same LET task should remain separated by the same task latency L_i otherwise the LET property is violated. This leads to the following definitions.

Definition 3 (Maximum offset-free MRT). *Given a task chain $\mathcal{T}$, we define maximum offset-free MRT of the chain*

$$L_{\mathcal{T}}^+ = \sup_{\substack{\phi_{\tau_i}^r \in \mathbb{Z}, \\ \phi_{\tau_i}^w = \phi_{\tau_i}^r + L_i}} L_{\mathcal{T}}(\phi_{\tau_i}^r, \phi_{\tau_i}^w). \quad (8)$$

Definition 4 (Minimum offset-free MRT). *Given a task chain $\mathcal{T}$, we define minimum offset-free MRT of the chain*

$$L_{\mathcal{T}}^- = \inf_{\substack{\phi_{\tau_i}^r \in \mathbb{Z}, \\ \phi_{\tau_i}^w = \phi_{\tau_i}^r + L_i}} L_{\mathcal{T}}(\phi_{\tau_i}^r, \phi_{\tau_i}^w). \quad (9)$$

The maximum offset-free MRT $L_{\mathcal{T}}^+$ represents the longest possible MRT among all external event times t_0 and all possible read offsets $\phi_{\tau_i}^r$. Hence, it is the safe bound on the reaction time regardless of the offset that tasks may have. The minimum offset-free MRT $L_{\mathcal{T}}^-$ represents how small the MRT may be upon a careful choice of the task offsets. The range from $L_{\mathcal{T}}^-$ to $L_{\mathcal{T}}^+$ represents the offset design space, which may be explored by designers. For example, Günzel et al. [5] have proposed an offset assignment method which reach the minimum offset-free MRT of $L_{\mathcal{T}}^-$, for some subclass of task periods (so called *max-harmonic*).

Figure 2 illustrates the task and chain model with an example, using the same job representation of Figure 1. The reaction time of the chain from two different instants t_0 and t'_0 is also represented.

IV. THE ZEROLET EQUIVALENT MODEL

All end-to-end latency metrics are the sum of two contributions: (i) the *intra-task* latency introduced by each task and (ii) the *inter-task* latency introduced by the synchronization between write and read events of tasks. In this section, we aim at decoupling these two distinct components. In the case of LET tasks, the intra-task latency is very simple: it is the sum of the separations $\phi_{\tau_i}^w - \phi_{\tau_i}^r$ between the read and the write offsets over all tasks. To focus on the inter-task latency only, in this section we introduce the zeroLET task model. As we show next, the zeroLET model enables a simpler but equivalent analysis of chains of LET tasks.

Definition 5 (zeroLET chain, zeroLET tasks). *Given a task chain $\mathcal{T} = [\tau_1, \dots, \tau_n]$, we define its zeroLET task chain $\mathcal{Z} = [\zeta_1, \dots, \zeta_n]$ as chain of zeroLET tasks ζ_i . Each zeroLET task $\zeta_i = \langle T_i, \phi_{\zeta_i} \rangle$ has the same period T_i as the corresponding task τ_i in $\mathcal{T}$, and the offset ϕ_{ζ_i} is defined recursively by*

$$\begin{cases} \phi_{\zeta_1} = \phi_{\tau_1}^r \\ \phi_{\zeta_{i+1}} = \phi_{\zeta_i} + \phi_{\tau_{i+1}}^r - \phi_{\tau_i}^w \end{cases} \quad (10)$$

Each task τ_i in $\mathcal{T}$ is replaced by a zeroLET task ζ_i (the “zeta” Greek letter) in $\mathcal{Z}$. A zeroLET task ζ_i has a logical

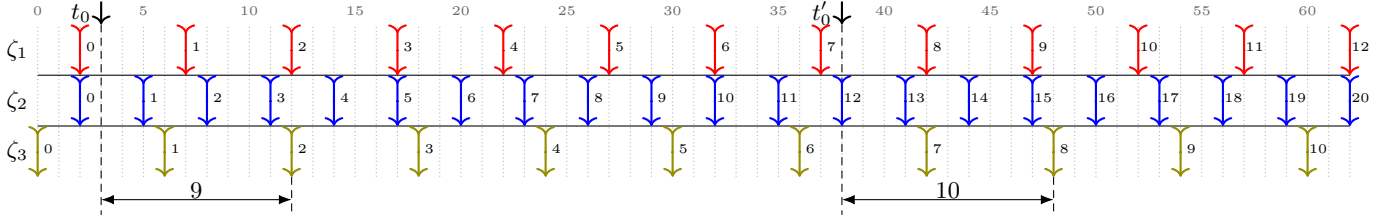


Fig. 3. Representing the zeroLET chain $\mathcal{Z} = [\zeta_1, \dots, \zeta_n]$ corresponding to the chain $\mathcal{T} = [\tau_1, \tau_2, \tau_3]$ of Figure 2. In this zeroLET chain, the reaction time of an external event occurring at time $t_0 = 3$ is $L_{\mathcal{Z}}(3, \phi_{\zeta_i}) = 9$, as depicted in the figure. As stated in Theorem 1, the reaction time $L_{\mathcal{T}}(3, \phi_{\tau_i}^r, \phi_{\tau_i}^w)$ from the same t_0 of the original chain $\mathcal{T}$ is $L_{\mathcal{Z}}(3, \phi_{\zeta_i}) + L_1 + L_2 + L_3 = 9 + 4 + 3 + 6 = 22$. Hence, to analyze the synchronization between tasks, it is sufficient to analyze the simpler zeroLET chain.

execution time equal to zero, that is the read and write instants of each job are coincident, formally,

$$\forall j \in \mathbb{Z}, \quad \text{rd}_{\zeta_i}(j) = \text{wr}_{\zeta_i}(j) = j T_i + \phi_{\zeta_i}. \quad (11)$$

The only offset ϕ_{ζ_i} is both the read and write offset of ζ_i .

The zeroLET chain is useful because it preserves the same separation between the write instant of any task and the read instants of the adjacent task in the chain. In fact, $\forall j_i, j_{i+1} \in \mathbb{Z}$ we have that

$$\begin{aligned} \text{rd}_{\tau_{i+1}}(j_{i+1}) - \text{wr}_{\tau_i}(j_i) &= j_{i+1} T_{i+1} + \phi_{\tau_{i+1}}^r - (j_i T_i + \phi_{\tau_i}^w) \\ &= j_{i+1} T_{i+1} - j_i T_i + \phi_{\tau_{i+1}}^r - \phi_{\tau_i}^w \\ &= j_{i+1} T_{i+1} - j_i T_i + \phi_{\zeta_{i+1}} - \phi_{\zeta_i} \\ &= \text{rd}_{\zeta_{i+1}}(j_{i+1}) - \text{wr}_{\zeta_i}(j_i) \end{aligned} \quad (12)$$

The zeroLET chains are a special class of generic LET chains, as defined earlier in Section III. As such, all definitions of Section III (Definitions 1, 2, 3, 4) are applicable to zeroLET chains.

Since zeroLET tasks only have a single offset ϕ_{ζ_i} and not the pair read/write offsets $(\phi_{\tau_i}^r, \phi_{\tau_i}^w)$ as the tasks τ_i in $\mathcal{T}$, the reaction time from some external event at time t_0 of the zeroLET chain $\mathcal{Z}$ can be more compactly written as

$$L_{\mathcal{Z}}(t_0, \phi_{\zeta_i}) = \text{wr}_{\zeta_n}(j_n) - t_0. \quad (13)$$

Notice that in the zeroLET chain $\mathcal{Z}$, the reaction time only depends on the offset ϕ_{ζ_i} of the zeroLET tasks, rather than on both read and write offsets.

Next theorem establishes a fundamental property of zeroLET chains, which justifies the interest in their usage.

Theorem 1. Given a task chain $\mathcal{T}$ and its zeroLET task chain $\mathcal{Z}$, for all event instants t_0 and all offsets, we have

$$L_{\mathcal{T}}(t_0, \phi_{\tau_i}^r, \phi_{\tau_i}^w) = L_{\mathcal{Z}}(t_0, \phi_{\zeta_i}) + \sum_{i=1}^n L_i. \quad (14)$$

Informally, the theorem states that the reaction time of any chain $\mathcal{T}$ can be computed from the reaction time of its zeroLET chain (inter-task latency) plus the latency L_i of all tasks τ_i (intra-task latency). The advantage of such expression is that $L_{\mathcal{Z}}(t_0, \phi_{\zeta_i})$ depends on fewer parameters than $L_{\mathcal{T}}(t_0, \phi_{\tau_i}^r, \phi_{\tau_i}^w)$ thus enabling a more effective design exploration or any other investigation or optimization.

Proof. For convenience, we recall that

$$L_{\mathcal{T}}(t_0, \phi_{\tau_i}^r, \phi_{\tau_i}^w) = \text{wr}_{\tau_n}(j_n) - t_0$$

with indices $(j_1, \dots, j_n)$ from Eq. (4), and

$$L_{\mathcal{Z}}(t_0, \phi_{\zeta_i}) = \text{wr}_{\zeta_n}(j'_n) - t_0$$

for possibly different job indices $(j'_1, \dots, j'_n)$ from the parameters of $\mathcal{Z}$.

We employ induction to prove that

$$\forall i = 1, \dots, n \quad j'_i = j_i.$$

First of all, from Def. 5 we know that all tasks in chain $\mathcal{T}$ have the same period of all corresponding tasks in the zeroLET chain $\mathcal{Z}$. Let us start proving $j'_i = j_i$ with case $i = 1$. And from Equations (10) and (4)

$$j'_1 = \left\lceil \frac{t_0 - \phi_{\zeta_1}}{T_1} \right\rceil = \left\lceil \frac{t_0 - \phi_{\tau_1}^r}{T_1} \right\rceil = j_1.$$

For the induction step with $i \geq 2$, we assume that $j'_i = j_i$ to be true and prove it for $j'_{i+1} = j_{i+1}$.

From Eq. (10) and Eq. (4), we have

$$\begin{aligned} j_{i+1} &= \left\lceil \frac{j_i T_i + \phi_{\tau_i}^w - \phi_{\tau_{i+1}}^r}{T_{i+1}} \right\rceil \\ &= \left\lceil \frac{j'_i T_i + \phi_{\zeta_i} - \phi_{\zeta_{i+1}}}{T_{i+1}} \right\rceil = j'_{i+1} \end{aligned}$$

We now evaluate $L_{\mathcal{T}}(t_0, \phi_{\tau_i}^r, \phi_{\tau_i}^w)$. By adding and subtracting the n write instants of the tasks in $\mathcal{T}$ we find

$$\begin{aligned} L_{\mathcal{T}}(t_0, \phi_{\tau_i}^r, \phi_{\tau_i}^w) &= \text{wr}_{\tau_n}(j_n) - t_0 \\ &= \text{wr}_{\tau_1}(j_1) - t_0 + \sum_{i=2}^n (\text{wr}_{\tau_i}(j_i) - \text{wr}_{\tau_{i-1}}(j_{i-1})) \end{aligned}$$

and by also adding and subtracting the n read instants

$$\begin{aligned} L_{\mathcal{T}}(t_0, \phi_{\tau_i}^r, \phi_{\tau_i}^w) &= \underbrace{\text{wr}_{\tau_1}(j_1) - \text{rd}_{\tau_1}(j_1)}_{\phi_{\tau_1}^w - \phi_{\tau_1}^r \text{ from Eq. (1)}} + \text{rd}_{\tau_1}(j_1) - t_0 \\ &\quad + \sum_{i=2}^n \underbrace{(\text{wr}_{\tau_i}(j_i) - \text{rd}_{\tau_i}(j_i))}_{\phi_{\tau_i}^w - \phi_{\tau_i}^r \text{ from Eq. (1)}} + \text{rd}_{\tau_i}(j_i) - \text{wr}_{\tau_{i-1}}(j_{i-1}) \\ &= \text{rd}_{\tau_1}(j_1) - t_0 + \sum_{i=1}^{n-1} (\text{rd}_{\tau_{i+1}}(j_{i+1}) - \text{wr}_{\tau_i}(j_i)) + \sum_{i=1}^n (\phi_{\tau_i}^w - \phi_{\tau_i}^r). \end{aligned}$$

We now derive the reaction time of the zeroLET chain. With similar steps as above,

$$\begin{aligned}
 L_{\mathcal{Z}}(t_0, \phi_{\zeta_i}) &= \text{wr}_{\zeta_n}(j_n) - t_0 \\
 &= \text{wr}_{\zeta_1}(j_1) - t_0 + \sum_{i=1}^{n-1} (\text{wr}_{\zeta_{i+1}}(j_{i+1}) - \text{wr}_{\zeta_i}(j_i)) \\
 &= \text{rd}_{\zeta_1}(j_1) - t_0 + \sum_{i=1}^{n-1} (\text{rd}_{\zeta_{i+1}}(j_{i+1}) - \text{wr}_{\zeta_i}(j_i)).
 \end{aligned}$$

In the last step, we use the coincidence of read with write instants for zeroLET tasks, i.e., $\text{rd}_{\zeta_i}(j) = \text{wr}_{\zeta_i}(j)$, as in Eq. (11).

We now observe that from Eq. (10), it follows $\text{rd}_{\tau_1}(j_1) = \text{rd}_{\zeta_1}(j_1)$. And finally, from Eq. (12) it follows that $L_{\mathcal{T}}(t_0, \phi_{\tau_i}^r, \phi_{\tau_i}^w)$ coincides with $L_{\mathcal{Z}}(t_0, \phi_{\zeta_i}) + \sum_{i=1}^n (\phi_{\tau_i}^w - \phi_{\tau_i}^r)$, hence concluding the proof. $\square$

Figure 3 illustrates the zeroLET chain $\mathcal{Z}$ corresponding to the chain $\mathcal{T}$ of the example in Figure 2. The chain $\mathcal{T}$ of Figure 2 is transformed into $\mathcal{Z} = [\langle 5, 2 \rangle, \langle 3, 2 \rangle, \langle 6, 0 \rangle]$, as indicated in Eq. (10) of Def. 5.

Theorem 1 demonstrates quite clearly the advantage of the zeroLET chains. The analysis of regular LET chains can be decoupled into two components:

- the synchronization among tasks, which can be analyzed by the zeroLET chain, and
- the intra-task latencies.

Such a theorem is stated for the reaction time $L_{\mathcal{T}}(t_0, \phi_{\tau_i}^r, \phi_{\tau_i}^w)$. The same composition property, however, is inherited by all other metrics: MRT of Def. 2, and the maximum (resp. minimum) offset-free MRT of Def. 3 (resp. Def. 4). In fact, we have

$$\begin{aligned}
 L_{\mathcal{T}}(\phi_{\tau_i}^r, \phi_{\tau_i}^w) &= L_{\mathcal{Z}}(\phi_{\zeta_i}) + \sum_{i=1}^n L_i \\
 L_{\mathcal{T}}^+ &= L_{\mathcal{Z}}^+ + \sum_{i=1}^n L_i \\
 L_{\mathcal{T}}^- &= L_{\mathcal{Z}}^- + \sum_{i=1}^n L_i
 \end{aligned}$$

with the following simpler definition on the zeroLET chain $\mathcal{Z}$

$$L_{\mathcal{Z}}(\phi_{\zeta_i}) = T_1 - 1 + \sup_{j_1 \in \mathbb{Z}} \{ \text{wr}_{\zeta_n}(j_n) - \text{rd}_{\zeta_1}(j_1) \} \quad (15)$$

$$L_{\mathcal{Z}}^+ = \sup_{\phi_{\zeta_i} \in \mathbb{Z}} L_{\mathcal{Z}}(\phi_{\zeta_i}) \quad (16)$$

$$L_{\mathcal{Z}}^- = \inf_{\phi_{\zeta_i} \in \mathbb{Z}} L_{\mathcal{Z}}(\phi_{\zeta_i}). \quad (17)$$

V. THE SPACE OF OFFSETS

In the definition of maximum offset-free MRT $L_{\mathcal{Z}}$ of Eq. (16), each task offsets ϕ_{ζ_i} spans over the whole $\mathbb{Z}$ as it is indeed required to evaluate the longest possible reaction time among all task offsets. Such a definition, however, is not very practical as the cardinality of the set for all ϕ_{ζ_i} is infinite. In the next paragraphs, we are determining some invariants of the function $L_{\mathcal{Z}}(\phi_{\zeta_i})$, which enables the calculation of $L_{\mathcal{Z}}^+$ and $L_{\mathcal{Z}}^-$ over a finite set.

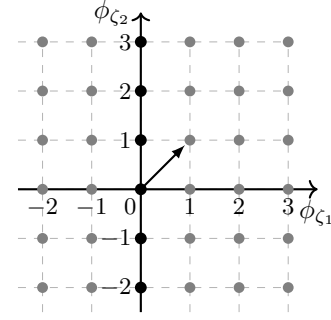


Fig. 4. By incrementing all offsets by one (represented by the diagonal arrow), the maximum reaction $L_{\mathcal{Z}}(\phi_{\zeta_i})$ does not vary. This means that for computing the maximum offset-free MRT of Eq. (16), we can evaluate $L_{\mathcal{Z}}(\phi_{\zeta_i})$ only when $\phi_{\zeta_1} = 0$ (black dots) and ignore all other offsets (gray dots).

Lemma 1. Given a zeroLET chain $\mathcal{Z}$, let us define the zeroLET chain $\mathcal{Z}'$ with

- same periods, and
- offsets equal to

$$\forall i = 1, \dots, n \quad \phi_{\zeta'_i} = \phi_{\zeta_i} + 1. \quad (18)$$

Then, the MRT $L_{\mathcal{Z}'}(\phi_{\zeta'_i})$ is

$$L_{\mathcal{Z}'}(\phi_{\zeta'_i}) = L_{\mathcal{Z}}(\phi_{\zeta_i}). \quad (19)$$

Proof. For convenience, we recall below the relation between job indices $(j_1, \dots, j_n)$ following from Eq. (4) and applied to the zeroLET task chain $\mathcal{Z}$, which is

$$j_{i+1} = \left\lceil \frac{j_i T_i + \phi_{\zeta_i} - \phi_{\zeta_{i+1}}}{T_{i+1}} \right\rceil, \quad i = 1, \dots, n-1.$$

We observe that by adding one to all offsets $(\phi_{\zeta_1}, \dots, \phi_{\zeta_n})$, the numerator of the expression above becomes

$$(\phi_{\zeta_i} + 1) - (\phi_{\zeta_{i+1}} + 1) = \phi_{\zeta_i} - \phi_{\zeta_{i+1}},$$

hence the indices $j_1, \dots, j_n$ do not change. Also, for these same indices we have

$$\begin{aligned}
 \text{wr}_{\zeta'_n}(j_n) - \text{rd}_{\zeta'_1}(j_1) &= \text{wr}_{\zeta_n}(j_n) + 1 - (\text{rd}_{\zeta_1}(j_1) + 1) = \\
 &= \text{wr}_{\zeta_n}(j_n) - \text{rd}_{\zeta_1}(j_1),
 \end{aligned}$$

from which we conclude that $L_{\mathcal{Z}'}(\phi_{\zeta'_i})$ as defined in Eq. (15), equals $L_{\mathcal{Z}}(\phi_{\zeta_i})$. This concludes the proof. $\square$

In words, $L_{\mathcal{Z}}(\phi_{\zeta_i})$ is invariant to a translation, which increments all offsets by the same amount.

Figure 4 illustrates Lemma 1. In the figure, an arrow represents the direction along which $L_{\mathcal{Z}}(\phi_{\zeta_i})$ is invariant. We remark that by Lemma 1 we can then shift all offsets by the same amount. For example, for the purpose of computing the supremum of (16), we can assume $\phi_{\zeta_1} = 0$ without loss of generality.

Next lemma establishes the invariance of $L_{\mathcal{Z}}(\phi_{\zeta_i})$ along other $n-1$ linearly independent directions. The combination of both lemmas will enable the computation of $L_{\mathcal{Z}}^+$ of Eq. (16), and of $L_{\mathcal{Z}}^-$ of Eq. (17) respectively, over a finite set.

Lemma 2. Given a zeroLET chain $\mathcal{Z}$, let us define G_i, H_i by
 $H_1 = T_1, H_i = \text{lcm}(H_{i-1}, T_i), G_i = \text{gcd}(H_{i-1}, T_i). \quad (20)$

Then

$$\forall k = 2, \dots, n, \quad L_{\mathcal{Z}}(\phi_{\zeta_i}) = L_{\mathcal{Z}'}(\phi_{\zeta'_i}) \quad (21)$$

with the zeroLET chain $\mathcal{Z}'$ defined by

- periods equal to $\mathcal{Z}$, and
- offset defined by

$$\begin{cases} \phi_{\zeta'_i} = \phi_{\zeta_i}, & i \leq k-1 \\ \phi_{\zeta'_k} = \phi_{\zeta_k} + G_k & i = k \\ \phi_{\zeta'_i} = \phi_{\zeta_i} + aH_{k-1} & i \geq k+1. \end{cases} \quad (22)$$

and with a being the coefficient of the following Bézout's identity.

$$aH_{k-1} = bT_k + G_k. \quad (23)$$

In words, we define H_i as the least common multiple of periods of all tasks up to ζ_i . Notice that in the vast literature on real-time systems, H_n is normally called the *hyperperiod* of the tasks. Given these definitions, the MRT function $L_{\mathcal{Z}}(\phi_{\zeta_i})$ is invariant by any of the $n-1$ translations of the offsets indicated in Eq. (22).

Proof. Let k be any task index from 2 to n , as in (21). The proof strategy is to show that for any job chain $(j_1, \dots, j_n)$ of $\mathcal{Z}$, we can always find a job chain $(j'_1, \dots, j'_n)$ of $\mathcal{Z}'$ such that

$$\text{wr}_{\zeta'_n}(j'_n) - \text{rd}_{\zeta'_1}(j'_1) = \text{wr}_{\zeta_n}(j_n) - \text{rd}_{\zeta_1}(j_1). \quad (24)$$

Also, we are showing that the found map $(j_1, \dots, j_n) \mapsto (j'_1, \dots, j'_n)$ is a bijection, implying then that the supremum of (15) can be made over $j'_1 \in \mathbb{Z}$.

Let us start by constructing j'_1 such that Eq. (24) holds. First, we observe that the coefficients of the Bézout's identity of (23) always exist because $G_k = \text{gcd}(H_{k-1}, T_k)$, by definition. We claim that the good choice for j'_1 is

$$j'_1 = j_1 + a \frac{H_{k-1}}{T_1}.$$

Let's prove that for such j'_1 Eq. (24) holds.

We proceed by proving by induction that

$$\forall i = 1, \dots, k-1, \quad j'_i = j_i + a \frac{H_{k-1}}{T_i},$$

with j'_i denoting the job index of ζ'_i found by $i-1$ applications of the ceiling function of

$$j'_{i+1} = \left\lceil \frac{j'_i T_i + \phi_{\zeta'_i} - \phi_{\zeta'_{i+1}}}{T_{i+1}} \right\rceil$$

used to define the read/write instant of job chains of $\mathcal{Z}'$.

The base case of the induction with $i = 1$ is given by our choice of j'_1 .

For the induction step, we assume the statement to be true for j'_i and proved it for j'_{i+1} . By definition, we have

$$j'_{i+1} = \left\lceil \frac{j'_i T_i + \phi_{\zeta'_i} - \phi_{\zeta'_{i+1}}}{T_{i+1}} \right\rceil$$

and by replacing the definition of j'_i , true for the inductive hypothesis, we get

$$j'_{i+1} = \left\lceil \frac{(j_i + a \frac{H_{k-1}}{T_i}) T_i + \phi_{\zeta'_i} - \phi_{\zeta'_{i+1}}}{T_{i+1}} \right\rceil$$

and with elementary algebra

$$j'_{i+1} = \left\lceil \frac{j_i T_i + \phi_{\zeta'_i} - \phi_{\zeta'_{i+1}}}{T_{i+1}} + a \frac{H_{k-1}}{T_{i+1}} \right\rceil.$$

Since H_{k-1} divides all periods from T_1 to T_{k-1} and $i+1 \leq k-1$, $\frac{H_{k-1}}{T_{i+1}}$ is integer and then it can be brought outside the ceiling operator, that is

$$j'_{i+1} = \left\lceil \frac{j_i T_i + \phi_{\zeta'_i} - \phi_{\zeta'_{i+1}}}{T_{i+1}} \right\rceil + a \frac{H_{k-1}}{T_{i+1}}.$$

From the definition $\phi_{\zeta'_i}$ with $i \leq k+1$ of Eq. (22)

$$j'_{i+1} = \left\lceil \frac{j_i T_i + \phi_{\zeta_i} - \phi_{\zeta_{i+1}}}{T_{i+1}} \right\rceil + a \frac{H_{k-1}}{T_{i+1}}$$

and by simply recognizing the definition of j_{i+1} from j_i , we conclude with

$$j'_{i+1} = j_{i+1} + a \frac{H_{k-1}}{T_{i+1}}.$$

The case for j'_k is special as we need to exploit the Bézout's identity of Eq. (23). We have

$$j'_k = \left\lceil \frac{j'_{k-1} T_{k-1} + \phi_{\zeta'_{k-1}} - \phi_{\zeta'_k}}{T_k} \right\rceil.$$

By replacing the just proved expression for j'_{k-1} and the definition of $\phi_{\zeta'_{k-1}}, \phi_{\zeta'_k}$ of Eq. (22), case $i = k$, we find

$$\begin{aligned} j'_k &= \left\lceil \frac{(j_{k-1} + a \frac{H_{k-1}}{T_{k-1}}) T_{k-1} + \phi_{\zeta_{k-1}} - \phi_{\zeta_k} - G_k}{T_k} \right\rceil \\ &= \left\lceil \frac{j_{k-1} T_{k-1} + a H_{k-1} + \phi_{\zeta_{k-1}} - \phi_{\zeta_k} - G_k}{T_k} \right\rceil. \end{aligned}$$

Now we invoke Bézout's identity of Eq. (23) and we find

$$\begin{aligned} j'_k &= \left\lceil \frac{j_{k-1} T_{k-1} + \phi_{\zeta_{k-1}} - \phi_{\zeta_k} + b T_k}{T_k} \right\rceil \\ &= \left\lceil \frac{j_{k-1} T_{k-1} + \phi_{\zeta_{k-1}} - \phi_{\zeta_k}}{T_k} + b \right\rceil = j_k + b. \end{aligned}$$

We are now proving by induction that

$$\forall i \geq k+1, \quad j'_i = j_i.$$

In the following base case, we use Bézout's identity Eq. (23) to evaluate j'_{k+1} :

$$\begin{aligned} j'_{k+1} &= \left\lceil \frac{j'_k T_k + \phi_{\zeta'_k} - \phi_{\zeta'_{k+1}}}{T_{k+1}} \right\rceil \\ &= \left\lceil \frac{(j_k + b) T_k + \phi_{\zeta_k} + G_k - (\phi_{\zeta_{k+1}} + a H_{k-1})}{T_{k+1}} \right\rceil \\ &= \left\lceil \frac{j_k T_k + \phi_{\zeta_k} - \phi_{\zeta_{k+1}}}{T_{k+1}} \right\rceil = j_{k+1}. \end{aligned}$$

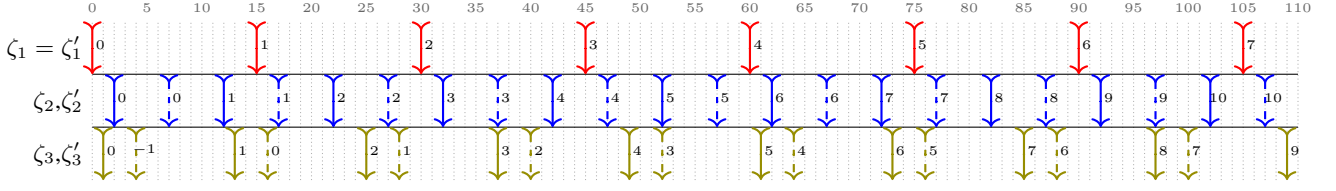


Fig. 5. Illustration of Lemma 2 with the chain $\mathcal{Z} = [(15, 0), (10, 2), (12, 1)]$. We also represent $\mathcal{Z}'$ by dashed arrows, for the case with $k = 2$. In such a case $H_1 = T_1 = 15$ and $T_2 = 10$. The coefficients of the Bézout's identity of Eq. (23) are $a = 1$ and $b = 1$, since $1 \times 15 = 1 \times 10 + 5$. The zeroLET chain $\mathcal{Z}'$ modified according to Eq. (22), has the following parameters $\mathcal{Z}' = [(15, 0), (10, 2 + 5), (12, 1 + 15)] = [(15, 0), (10, 7), (12, 16)]$. The figure also illustrate the mapping between job chains of $\mathcal{Z}$ and $\mathcal{Z}'$ with the same reaction time. For example, the job chain $(j_1, j_2, j_3) = (4, 6, 6)$ of $\mathcal{Z}$ is mapped to the job $(j'_1, j'_2, j'_3) = (5, 7, 6)$ of $\mathcal{Z}'$.

For the induction step with $i + 1 \geq k + 1$,

$$j'_{i+1} = \left\lfloor \frac{j'_i T_i + \phi_{\zeta'_i} - \phi_{\zeta'_{i+1}}}{T_{i+1}} \right\rfloor = \left\lfloor \frac{j_i T_i + \phi_{\zeta_i} - \phi_{\zeta_{i+1}}}{T_{i+1}} \right\rfloor = j_{i+1}$$

as required.

We remark that the map

$$\begin{cases} j_i \mapsto j'_i = j_i + a \frac{H_{k-1}}{T_i}, & i \leq k-1 \\ j_k \mapsto j'_k = j_k + b \\ j_i \mapsto j'_i = j_i & i \geq k+1 \end{cases}$$

is a bijection as we are simply adding a vector to $(j_1, \dots, j_n)$.

We now need to prove that such a map preserves the reaction time, that is Eq. (24) holds. In fact

$$\begin{aligned} \text{rd}_{\zeta'_1}(j'_1) &= j'_1 T_1 + \phi_{\zeta'_1} = (j_1 + a \frac{H_{k-1}}{T_1}) T_1 + \phi_{\zeta_1} \\ &= \text{rd}_{\zeta_1}(j_1) + a H_{k-1}. \end{aligned}$$

To determine $\text{wr}_{\zeta'_n}(j'_n)$ we need to consider two cases. If $k = n$ then

$$\begin{aligned} \text{wr}_{\zeta'_n}(j'_n) &= j'_n T_n + \phi_{\zeta'_n} = (j_n + b) T_n + \phi_{\zeta_n} + G_n \\ &= \text{wr}_{\zeta_n}(j_n) + b T_n + G_n = \text{wr}_{\zeta_n}(j_n) + b T_k + G_k \end{aligned}$$

If instead $k < n$ then

$$\begin{aligned} \text{wr}_{\zeta'_n}(j'_n) &= j'_n T_n + \phi_{\zeta'_n} = j_n T_n + \phi_{\zeta_n} + a H_{k-1} \\ &= \text{wr}_{\zeta_n}(j_n) + b T_k + G_k, \end{aligned}$$

which is the same as the case with $k = n$.

In conclusion

$$\begin{aligned} \text{wr}_{\zeta'_n}(j'_n) - \text{rd}_{\zeta'_1}(j'_1) &= \\ &= \text{wr}_{\zeta_n}(j_n) + b T_k + G_k - \text{rd}_{\zeta_1}(j_1) - a H_{k-1} \end{aligned}$$

which is equal to $\text{wr}_{\zeta_n}(j_n) - \text{rd}_{\zeta_1}(j_1)$ because of Eq. (23). We have then proved (24) and from such property it holds that

$$\sup_{j'_1 \in \mathbb{Z}} \{ \text{wr}_{\zeta'_n}(j'_n) - \text{rd}_{\zeta'_1}(j'_1) \} = \sup_{j_1 \in \mathbb{Z}} \{ \text{wr}_{\zeta_n}(j_n) - \text{rd}_{\zeta_1}(j_1) \}.$$

and from the definition of MRT $L_{\mathcal{Z}}(\phi_{\zeta_i})$ of Eq. (15), it immediately follows that

$$L_{\mathcal{Z}'}(\phi_{\zeta'_i}) = L_{\mathcal{Z}}(\phi_{\zeta_i}),$$

as required. $\square$

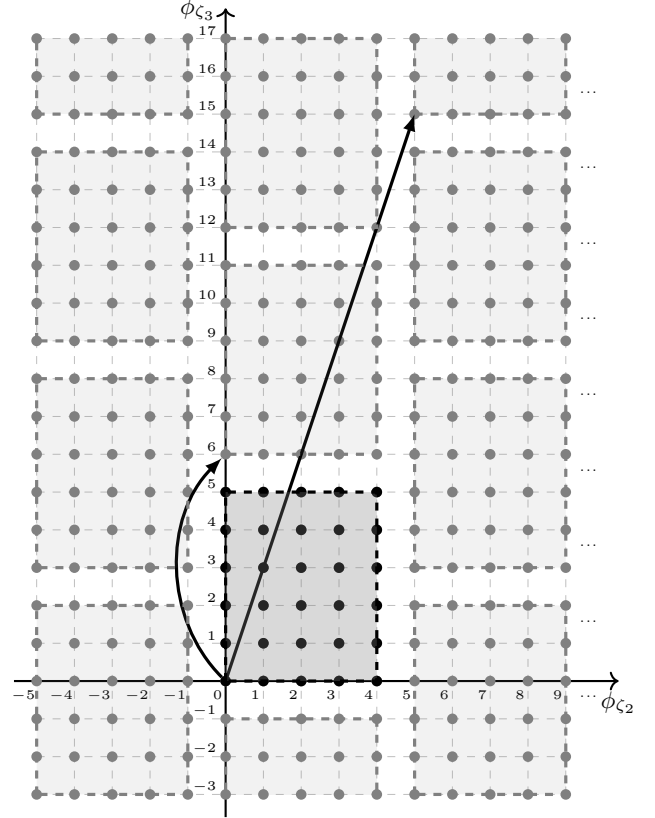


Fig. 6. Lemma 2 demonstrates the invariance of $L_{\mathcal{Z}}(\phi_{\zeta_i})$ along the $n - 1$ offset translation given by Eq. (22) for $k = 2, \dots, n$. Here we illustrate these vectors when $T_1 = 15$, $T_2 = 10$ and $T_3 = 12$, which are the same time periods of the chain illustrated in Figure 5. For this example, when $k = 2$, the Bézout's identity of (23) is $1 \times 15 = 1 \times 10 + 5$ and then from Eq. (22) by incrementing all offsets by $(0, 5, 15)$ the MRT $L_{\mathcal{Z}}(\phi_{\zeta_i})$ does not change (showing in the figure only the projection over the $(\phi_{\zeta_2}, \phi_{\zeta_3})$ plane). When $k = 3$, Lemma 2 demonstrates that the MRT $L_{\mathcal{Z}}(\phi_{\zeta_i})$ is invariant with the offset increment $(0, 0, 6)$. The coverage of the whole $\mathbb{Z}^2$ by Φ , represented by a rectangle with black dashed boundary, and its translations by Eq. (22) is the graphical representation of Corollary 1. The values of $L_{\mathcal{Z}}(\phi_{\zeta_i})$ for all offsets in the black rectangle are reported in Table II.

Lemma 1 and Lemma 2 establish very well defined n patterns of repetition of $L_{\mathcal{Z}}(\phi_{\zeta_i})$. Specifically, if the offsets are changed by any integer combination of Eq. (18) and Eq. (22) for any $k = 2, \dots, n$, then the maximum reaction time $L_{\mathcal{Z}}(\phi_{\zeta_i})$ does not change.

Corollary 1. Given a zeroLET task chain $\mathcal{Z}$, the maximum

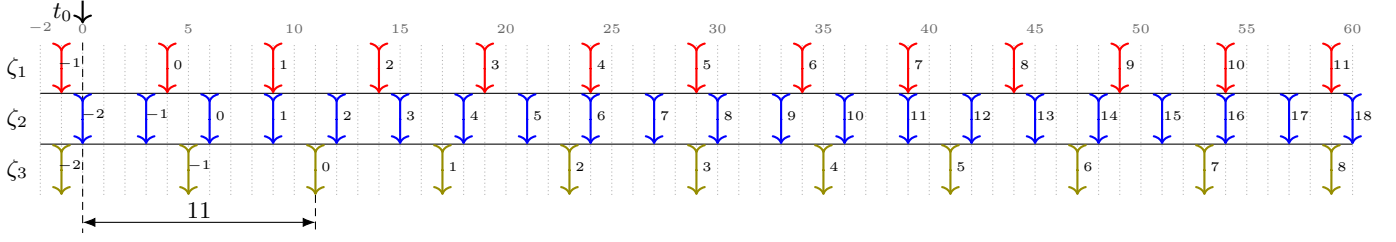


Fig. 7. Representing the zeroLET chain $\mathcal{Z} = \langle [5, 4], [3, 6], [6, 11] \rangle$ which has the same periods of Figure 3. In this zeroLET chain, the reaction time of an external event occurring at time $t_0 = 0$ is $L_{\mathcal{Z}}(0, \phi_{\zeta_1}) = 11$, as depicted in the figure. It is also the maximum offset-free MRT $L_{\mathcal{Z}}^+ = \sum_{i=1}^3 T_i - 3 = 5 + 3 + 6 - 3 = 11$ with offsets $\phi_{\zeta_1} = T_1 - 1 = 4$, $\phi_{\zeta_2} = T_1 + T_2 - 2 = 6$, $\phi_{\zeta_3} = T_1 + T_2 + T_3 - 3 = 11$, which conforms to Theorem 2.

and minimum offset-free MRTs are equal to

$$L_{\mathcal{Z}}^+ = \max_{(\phi_{\zeta_1}, \dots, \phi_{\zeta_n}) \in \Phi} L_{\mathcal{Z}}(\phi_{\zeta_i}) \quad (25)$$

$$L_{\mathcal{Z}}^- = \min_{(\phi_{\zeta_1}, \dots, \phi_{\zeta_n}) \in \Phi} L_{\mathcal{Z}}(\phi_{\zeta_i}) \quad (26)$$

with

$$\Phi = \{(\phi_{\zeta_1}, \dots, \phi_{\zeta_n}) \in \mathbb{Z}^n : \phi_{\zeta_1} = 0, 0 \leq \phi_{\zeta_i} \leq G_i - 1\} \quad (27)$$

and G_i defined by Eq. (20).

Proof. The proof strategy is to show that any value that $L_{\mathcal{Z}}(\phi_{\zeta_i})$ may take is also taken over the set Φ . The intuition of the corollary is also illustrated in Figure 6, where it is shown that the set Φ can cover the full $\mathbb{Z}^n$ by the translations of Lemma 2.

Formally, for any $(\phi_{\zeta_1}, \dots, \phi_{\zeta_n}) \in \mathbb{Z}^n$, from Lemma 1 we can decrease all offsets by ϕ_{ζ_1} without changing $L_{\mathcal{Z}}(\phi_{\zeta_i})$. Hence, we can restrict to offsets $(0, \phi_{\zeta_2}, \dots, \phi_{\zeta_n})$.

Let us now write $\phi_{\zeta_2} = q_2 G_2 + r_2$, by making the integer division of ϕ_{ζ_2} by G_2 . Lemma 2 guarantees that we can apply q_2 times the translation of Eq. (22) when $k = 2$, without changing the MRT $L_{\mathcal{Z}}(\phi_{\zeta_i})$. Such a transformation maps $\phi_{\zeta_2} \mapsto r_2$, with $0 \leq r_2 \leq G_2 - 1$ since r_2 is the remainder of an integer division by G_2 . Observing that the translation of Eq. (22) with any k does not change the offset ϕ_{ζ_i} for all $i \leq k - 1$, we can apply all translations of from k down to n and finally get some equivalent offset $(0, r_2, \dots, r_n)$ which belongs by construction to Φ defined in Eq. (27). The statement of Equations (25) and (26) then follow by observing that the supremum (respectively, infimum) over the finite set Φ indeed becomes a maximum (resp. minimum). $\square$

Corollary 1 enables an evaluation of the reaction time over a finite set. It is then interesting to determine the complexity of such an evaluation. From Equations (7) and (25), the maximum offset-free MRT can be computed by

$$L_{\mathcal{Z}}^+ = \max_{(\phi_{\zeta_i})_i \in \Phi} \max_{j_1 \in [0, \frac{H_1}{T_1}]} \{wr_{\zeta_n}(j_n) - rd_{\zeta_1}(j_1)\} + T_1 - 1.$$

From Eq. (4), the complexity for computing $wr_{\zeta_n}(j_n)$ is $O(n)$. The complexity of the set Φ is

$$|\Phi| = G_2 \cdot G_3 \cdots G_n = \frac{H_1 T_2}{H_2} \frac{H_2 T_3}{H_3} \cdots \frac{H_{n-1} T_n}{H_n} = \frac{1}{H_n} \prod_{i=1}^n T_i.$$

TABLE II

$L_{\mathcal{Z}}(\phi_{\zeta_i})$, WHEN $T_1 = 15, T_2 = 10, T_3 = 12$, AND $\phi_{\zeta_1} = 0$

$\phi_{\zeta_3} \setminus \phi_{\zeta_2}$	0	1	2	3	4
0	29	29	29	29	29
1	30	30	30	30	30
2	25	31	31	31	31
3	26	26	32	32	32
4	27	27	27	33	33
5	28	28	28	28	34

By multiplying with the cardinality of the set for j_1 and the complexity $O(n)$ for computing $wr_{\zeta_n}(j_n)$, we finally get that the complexity C for computing $L_{\mathcal{Z}}$ and $L_{\mathcal{Z}}^+$ is

$$C = n \prod_{i=2}^n T_i. \quad (28)$$

Such a complexity factory C will be later used in the evaluation of Section VI. We also observe that the cardinality of Φ is 1 when all periods are co-prime. In such a case, from Corollary 1 it follows immediately that $L_{\mathcal{Z}}^+ = L_{\mathcal{Z}}^-$ because they are maximum and minimum of the same function over a set of cardinality 1.

Table II shows an evaluation of $L_{\mathcal{Z}}(\phi_{\zeta_i})$ for the choice of $T_1 = 15, T_2 = 10$, and $T_3 = 12$, the same example used in Figure 6 to illustrate the coverage of all possible offsets by translations of Φ of Eq. (27).

The complexity of Eq. (28) for computing $L_{\mathcal{Z}}^+$ and $L_{\mathcal{Z}}^-$ may grow significantly with the length of the chain. Fortunately, the following theorem determines the value of $L_{\mathcal{Z}}^+$ in linear time.

Theorem 2. Given a zeroLET task chain $\mathcal{Z}$, the maximum offset-free MRT of

$$L_{\mathcal{Z}}^+ = \sum_{i=1}^n T_i - n \quad (29)$$

and it is taken when the offsets are

$$\forall i \in 1, \dots, n \quad \phi_{\zeta_i} = \sum_{k=1}^i T_k - i \quad (30)$$

Proof. We first prove that $\sum_{i=1}^n T_i - n$ is an upper bound for $L_{\mathcal{Z}}^+$ and then show that the offsets of Eq. (30) yields such value.

We want to prove that for any given $j_1 \in \mathbb{Z}$, the following property holds

$$\forall i = 2, \dots, n, \quad j_i \leq \frac{j_1 T_1 + \phi_{\zeta_1} - \phi_{\zeta_i} + \sum_{k=2}^i T_k - i + 1}{T_i} \quad (31)$$

with j_i defined from j_1 through $i - 1$ applications of Eq. (4) as needed for the definition of reaction time.

We recall first the following property of the ceiling function over rational numbers

$$\forall p \in \mathbb{Z}, q \in \mathbb{N}, \quad \left\lceil \frac{p}{q} \right\rceil \leq \frac{p + q - 1}{q} \quad (32)$$

which is needed to demonstrate Eq. (31).

We now prove the base case of Eq. (31). When $i = 2$ then we find

$$j_2 = \left\lceil \frac{j_1 T_1 + \phi_{\zeta_1} - \phi_{\zeta_2}}{T_2} \right\rceil \leq \frac{j_1 T_1 + \phi_{\zeta_1} - \phi_{\zeta_2} + T_2 - 1}{T_2}$$

by applying the property of (32). This proves the base case.

For the inductive step, we aim at proving Eq. (31) for $i + 1$ by assuming it true for i . We have

$$\begin{aligned} j_{i+1} &= \left\lceil \frac{j_i T_i + \phi_{\zeta_i} - \phi_{\zeta_{i+1}}}{T_{i+1}} \right\rceil \leq \frac{j_i T_i + \phi_{\zeta_i} - \phi_{\zeta_{i+1}} + T_{i+1} - 1}{T_{i+1}} \\ &\leq \frac{\frac{j_1 T_1 + \phi_{\zeta_1} - \phi_{\zeta_i} + \sum_{k=2}^i T_k - i + 1}{T_i} T_i + \phi_{\zeta_i} - \phi_{\zeta_{i+1}} + T_{i+1} - 1}{T_{i+1}} \\ &= \frac{j_1 T_1 + \phi_{\zeta_1} - \phi_{\zeta_{i+1}} + \sum_{k=2}^{i+1} T_k - i}{T_{i+1}}, \end{aligned}$$

which is exactly the statement of Eq. (31) for $i + 1$.

We are now exploiting the bound on j_n of Eq. (31) to determine a bound on the MRT $L_{\mathcal{Z}}(\phi_{\zeta_i})$ of Eq. (15). We start by bounding the “difficult” piece of $L_{\mathcal{Z}}(\phi_{\zeta_i})$ below,

$$\begin{aligned} \sup_{j_1 \in \mathbb{Z}} \{ \text{wr}_{\zeta_n}(j_n) - \text{rd}_{\zeta_1}(j_1) \} &= \sup_{j_1 \in \mathbb{Z}} \{ j_n T_n + \phi_{\zeta_n} - j_1 T_1 - \phi_{\zeta_1} \} \\ &\leq \sup_{j_1 \in \mathbb{Z}} \{ j_1 T_1 + \phi_{\zeta_1} - \phi_{\zeta_n} + \sum_{k=2}^n T_k - n + 1 + \phi_{\zeta_n} - j_1 T_1 - \phi_{\zeta_1} \} \\ &= \sum_{k=2}^n T_k - n + 1. \end{aligned}$$

By recalling now the definition of $L_{\mathcal{Z}}(\phi_{\zeta_i})$ of Eq. (15), we immediately get

$$L_{\mathcal{Z}}(\phi_{\zeta_i}) \leq \sum_{k=1}^n T_k - n.$$

We are now demonstrating that with the offset assignment of (30), the reaction time $L_{\mathcal{Z}}(0, \phi_{\zeta_i})$ of an external event at time $t_0 = 0$, is exactly equal to the upper bound above $\sum_{k=1}^n T_k - n$. We start by showing the job indices j_i of (4) from $t_0 = 0$ are

$$\forall i = 1, \dots, n, \quad j_i = 0.$$

TABLE III
EXAMPLE OF $L_{\mathcal{Z}}(\phi_{\zeta_i})$ FOR ALL $(\phi_{\zeta_i}) \in \Phi$, WHEN $T_1 = 12, T_2 = 8, T_3 = 9$. SHOWING ONLY $(\phi_{\zeta_2}, \phi_{\zeta_3})$ BECAUSE $\phi_{\zeta_1} = 0$.

$\phi_{\zeta_3} \setminus \phi_{\zeta_2}$	0	1	2	3
0	23	23	23	26
1	21	24	24	24
2	22	22	25	25

We prove it, again, by induction. Before applying the induction argument, we substitute Eq. (30) into the expression under consideration. For the base case

$$j_1 = \left\lceil \frac{t_0 - \phi_{\zeta_1}}{T_1} \right\rceil = \left\lceil \frac{0 - T_1 + 1}{T_1} \right\rceil = \left\lceil \frac{1}{T_1} \right\rceil - 1 = 0.$$

Assuming $j_i = 0$, we find

$$j_{i+1} = \left\lceil \frac{j_i T_i + \phi_{\zeta_i} - \phi_{\zeta_{i+1}}}{T_{i+1}} \right\rceil = \left\lceil \frac{0 - T_{i+1} + 1}{T_{i+1}} \right\rceil = 0.$$

Hence, from Eq. (13), the reaction time from $t_0 = 0$ is

$$L_{\mathcal{Z}}(0, \phi_{\zeta_i}) = \text{wr}_{\zeta_n}(0) - 0 = \phi_{\zeta_n} = \sum_{i=1}^n T_i - n.$$

In conclusion,

$$\sum_{i=1}^n T_i - n = L_{\mathcal{Z}}(0, \phi_{\zeta_i}) \leq L_{\mathcal{Z}}^+ = \sup_{\phi_{\zeta_i} \in \Phi} L_{\mathcal{Z}}(\phi_{\zeta_i}) \leq \sum_{i=1}^n T_i - n$$

which finally demonstrates Eq. (29). $\square$

We remark that the maximum offset-free MRT of Eq. (29) coincides with the ones given by Davare et al. [7, Section 2.1] and by Günzel et al. [9], after a proper adaptation of the notation and by assuming integer times. The added value of Theorem 2 is the demonstration that such a bound is tight by exhibiting the concrete offset assignment of Eq. (30).

Let us now illustrate the results found thus far with the example of chain $\mathcal{Z} = [\langle 12, 0 \rangle, \langle 8, \phi_{\zeta_2} \rangle, \langle 9, \phi_{\zeta_3} \rangle]$ with unknown offsets ϕ_{ζ_2} and ϕ_{ζ_3} . From Lemma 1, we can assume $\phi_{\zeta_1} = 0$ without loss of generality. Table III shows the values of $L_{\mathcal{Z}}(\phi_{\zeta_i})$ for all offsets in Φ , as defined by Eq. (27). The red numbers in the table represent the maximum offset-free MRT, and the blue numbers represent the minimum offset-free MRT. With this example, from Eq. (29), we can derive the maximum offset-free MRT $L_{\mathcal{Z}}^+ = \sum_{i=1}^3 T_i - 3 = 26$. From Eq. (30), such a maximum occurs with offsets $(\phi_{\zeta_1}, \phi_{\zeta_2}, \phi_{\zeta_3}) = (11, 18, 26)$. Lemma 1 states that we can add or subtract a common quantity from all offsets. By subtracting 11, we find the offsets $(\phi_{\zeta_1}, \phi_{\zeta_2}, \phi_{\zeta_3}) = (0, 7, 15)$ which still yields $L_{\mathcal{Z}}^+ = 26$. To find the equivalent offset ϕ_{ζ_2} in Φ we use Lemma 2 with $k = 2$. With $H_1 = T_1 = 12$ and $G_2 = \gcd(H_1, T_2) = 4$, the Bézout's identity of Eq. (23) is $1 \times 12 = 1 \times 8 + 4$, hence with $a = 1$. By applying the offset shift of Eq. (22), we get another equivalent offset of $(\phi_{\zeta_1}, \phi_{\zeta_2}, \phi_{\zeta_3}) = (0, 7, 15) - (0, 4, 12) = (0, 3, 3)$. Finally, by applying 4 times the shift of Lemma (22) with $k = 3$ and then with $G_3 = 3$, we find $(\phi_{\zeta_1}, \phi_{\zeta_2}, \phi_{\zeta_3}) = (0, 3, 0)$, which is the offset value shown in red the table.

TABLE IV
SELECTED SAMPLED PERIODS AND CORRESPONDING
MAXIMUM/MINIMUM OFFSET-FREE MRT.

$[T_i]$	$L_{\mathcal{Z}}^-$	ϕ_{ζ_i} of $L_{\mathcal{Z}}^-$	$L_{\mathcal{Z}}^+$	ϕ_{ζ_i} of $L_{\mathcal{Z}}^+$	$L_{\mathcal{Z}}^+ - L_{\mathcal{Z}}^-$
[12, 12, 12]	11	[0, 0, 0]	33	[0, 11, 10]	22
[10, 10, 10]	9	[0, 0, 0]	27	[0, 9, 8]	18
[9, 10, 11]	27	[0, 0, 0]	27	[0, 0, 0]	0
[8, 12, 12]	15	[0, 0, 0]	29	[0, 3, 2]	14
[12, 9, 12]	17	[0, 0, 6]	30	[0, 2, 7]	13
[10, 12, 12]	19	[0, 0, 0]	31	[0, 1, 0]	12
[5, 12, 12]	15	[0, 0, 0]	26	[0, 0, 11]	11
[9, 6, 9]	11	[0, 0, 3]	21	[0, 2, 4]	10
[10, 3, 10]	11	[0, 0, 2]	20	[0, 0, 1]	9
[9, 7, 9]	14	[0, 0, 6]	22	[0, 0, 5]	8
[6, 6, 9]	11	[0, 0, 0]	18	[0, 5, 1]	7
[4, 8, 12]	15	[0, 0, 0]	21	[0, 3, 2]	6
[5, 6, 10]	13	[0, 0, 4]	18	[0, 0, 9]	5
[3, 6, 9]	11	[0, 0, 0]	15	[0, 2, 1]	4
[2, 9, 12]	17	[0, 0, 0]	20	[0, 0, 5]	3
[3, 5, 3]	6	[0, 0, 1]	8	[0, 0, 0]	2
[2, 4, 3]	5	[0, 0, 0]	6	[0, 1, 0]	1
[2, 3, 5]	7	[0, 0, 0]	7	[0, 0, 0]	0

VI. EVALUATION

In this section, we illustrate an evaluation of the offset space, which is made simpler with the zeroLET model.

In a first set of experiments, we have been spanning the offset space for all chains of $n = 3$ tasks with periods from 2 to 12. Table IV lists a sample of the evaluated periods.¹ In the table, the maximum and minimum offset-free MRTs are written in blue and red, respectively. For both of them, we give the task offset in Φ (defined in Eq. (27)) giving such a value. The last column reports the difference between the maximum and minimum offset-free MRTs, which represents the breadth of the offset design space for the given choice of period.

Among the tested periods, the broadest offset space occurs when all periods are equal to 12 (the maximum period in the tested range). In fact, when periods are all the same equal to a common T , it is always $L_{\mathcal{Z}}^- = T - 1$ and $L_{\mathcal{Z}}^+ = n(T - 1)$. We would like to comment explicitly the values of $L_{\mathcal{Z}}^-$ and $L_{\mathcal{Z}}^+$ for the periods [10, 10, 10] and [9, 10, 11]. Their sum is the same, then from Eq. (29), the two values of $L_{\mathcal{Z}}^+$ coincide. These samples, however, differ in harmonicity with one triplet having $\gcd(T_i) = T_i$ and the other with all co-prime periods. In fact $L_{\mathcal{Z}}^-$, which is very sensitive to harmonicity, is 9 for periods [10, 10, 10] and $27 = L_{\mathcal{Z}}^+$ for [9, 10, 11]. On the one hand, the designer may be willing to choose periods [10, 10, 10] and set the offsets to achieve a MRT equal to $L_{\mathcal{Z}}^-$. On the other hand, such a choice may be very sensitive to offset variations and could result in a significant increase of the MRT $L_{\mathcal{Z}}(\phi_{\zeta_i})$ if the offsets are not carefully controlled. A second observation from Table IV is that the offset [0, 0, 0] seems to often produce a low value of $L_{\mathcal{Z}}(\phi_{\zeta_i})$. This is not true in general. Instead, it actually happens that the offset [0, 0, 0] yields the maximum offset-free MRT, as when periods are [3, 5, 3].

The margin in the reaction time offered by harmonic periods is a well known fact. Table V shows the “extreme” case of all task periods equal to 8. Table VI shows a case of

TABLE V
EXAMPLE OF $L_{\mathcal{Z}}(\phi_{\zeta_i})$, WHEN $T_1 = 8, T_2 = 8, T_3 = 8$, AND $\phi_{\zeta_1} = 0$

$\phi_{\zeta_3} \setminus \phi_{\zeta_2}$	0	1	2	3	4	5	6	7
0	7	15	15	15	15	15	15	15
1	8	8	16	16	16	16	16	16
2	9	9	9	17	17	17	17	17
3	10	10	10	10	18	18	18	18
4	11	11	11	11	11	19	19	19
5	12	12	12	12	12	12	20	20
6	13	13	13	13	13	13	13	21
7	14	14	14	14	14	14	14	14

TABLE VI
EXAMPLE OF $L_{\mathcal{Z}}(\phi_{\zeta_i})$, WHEN $T_1 = 4, T_2 = 12, T_3 = 6$, AND $\phi_{\zeta_1} = 0$

$\phi_{\zeta_3} \setminus \phi_{\zeta_2}$	0	1	2	3
0	11	17	17	17
1	12	12	18	18
2	13	13	13	19
3	14	14	14	14
4	15	15	15	15
5	16	16	16	16

“max-harmonic” periods: all task periods divide the maximum. When periods fall into these categories, Günzel and Becker [5] proposed an offset assignment which yields the minimum offset-free MRT. Their offset assignment and the corresponding MRT $L_{\mathcal{Z}}(\phi_{\zeta_i})$ are coherent with the results shown in the tables, provided that they are adapted to the integer nature of time.

This confirms that for max-harmonic systems, zeroLET provides an equivalent characterization while avoiding explicit offset construction. Furthermore, our analysis can satisfy general cases and reduce the computational cost of exploring the offset space.

Now, we evaluate how the runtime of the offset exploration is affected by the number of tasks in the chain and its relation with the task periods. We randomly generated chains with length ranging from $n = 3$ to $n = 10$, with 100 instances for each chain length. Following the automotive benchmark proposed by Kramer et al. [40], the period of each task was selected from $\{1, 2, 5, 10, 20, 50, 100, 200, 500, 1000\}$. For each generated chain, we exhaustively span the offset space Φ of Eq. (27) to compute the maximum and minimum offset-free MRTs.²

We denote the complexity expression in Eq. (28) by C and the measured runtime by R . Figure 8 shows the relationship between these two quantities. In general, R increases with C . The points corresponding to different chain lengths overlap because chain length alone does not determine the runtime. In fact, a shorter chain with large periods may be more expensive to analyze than a longer chain with smaller periods. The runtime of the design space exploration is determined jointly by the chain length and the size of the search space; in particular, large period values may lead to a very broad search space, as indicated by Eq. (28). With the above settings, as the number of tasks grows, the offset space exploration for an

¹Note to the reviewers: if the paper is accepted, the full evaluation of min/max offset-free MRT and the corresponding best/worst case offsets will be made available on a public git repository.

²Note to the reviewers: if the paper is accepted, the C code for offset space exploration will be made available on a public git repository.

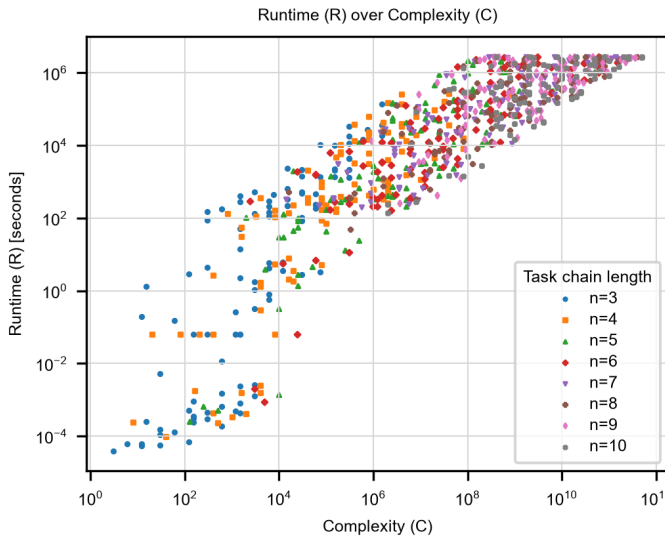


Fig. 8. Relationship between the complexity C , as defined by Eq. (28) and the runtime R . Both axes are in log scale.

increasing number of task sets was taking a too long runtime. For this reason, we discarded task sets with a complexity factor C larger than $C_{\max} = 10^{12}$.

VII. CONCLUSIONS

We propose the zeroLET task model to decouple the synchronization latency and the task latency. We use such a model to define equivalent classes of offsets, all yielding the same MRT. The quotient set of these equivalence can then be used to explore the design space of offset more efficiently. Also, we defined the maximum/minimum offset-free MRT to evaluate the breadth of such a design space.

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